SURFACE GRAVITY WAVES AND INTERNAL WAVES

Concepts, restoring forces, propagation scales, stratification and observational evidence

Original question

Briefly explain the basic concepts of ocean surface gravity waves and internal waves. Identify at least three differences between them in terms of where they occur, their restoring forces, propagation speeds, and amplitudes. Explain why internal waves can occur in a density-stratified ocean, and describe the differences between interfacial internal waves and internal waves in a continuously stratified ocean.

ABSTRACT

Surface gravity waves displace the air-sea interface and are restored mainly by gravity, whereas internal waves displace density surfaces inside a stably stratified ocean and are restored by buoyancy. This report compares their locations, restoring forces, propagation speeds, amplitudes and observable signatures; explains the stability condition $N^2>0$; and distinguishes a two-layer interfacial model from vertical modes in continuous stratification. A 10 s deep-water surface wave has a phase speed of 15.61 m s^{-1}, compared with 0.85 m s^{-1} for the worked two-layer internal-wave example. NOAA buoy observations document the surface-wave field, while a quality-controlled South China Sea Argo profile shows stable density stratification capable of supporting internal waves. The profile is interpreted as a diagnostic of wave-supporting conditions rather than direct evidence of a propagating internal wave.

DATA	METHODS	VALIDATION
NOAA NDBC 44088, 2024	Wave statistics	Independent MATLAB checks
Argo 2902695, 16 Mar 2017	TEOS-10 density and N^2	Reproducible source files

1. Core concepts and direct comparison

Surface gravity waves are disturbances of the air-sea interface. Water parcels mainly move in orbital paths while the wave form and energy propagate. Gravity acting on the displaced free surface is the restoring force. **Internal waves** are oscillations of density surfaces within a stably stratified water column. Buoyancy generated by gravity acting on internal density differences is their restoring force.

Feature	Surface gravity waves	Internal waves
Location	Free surface; motion usually decays downward.	Inside a stratified water column, often near a pycnocline.
Restoring force	Gravity acting across the large air-water density jump.	Buoyancy associated with smaller density differences in seawater.
Speed	Generally faster; depends on period/wavelength and depth.	Generally slower because reduced gravity is small.
Amplitude	Surface elevation commonly centimetres to several metres.	Isopycnal displacement commonly metres to tens of metres and sometimes >100 m.
Visibility	Directly visible and measured by buoys/altimeters.	Often hidden; detected by density, velocity or surface roughness signatures.

Wave-speed examples

Deep-water surface waves: $\omega^2=gk$, so $c_p=\omega/k=gT/(2\pi)$. For T=10 s, $c_p=15.61$ m s^{-1} and wavelength $\lambda=156.13$ m. Shallow-water surface waves have $c=(gH)^{1/2}$.

Two-layer internal long waves: $c=[g'h_1h_2/(h_1+h_2)]^{1/2}$, with $g'=g\Delta\rho/\rho_0$. For $h_1=50$ m, $h_2=150$ m, $\Delta\rho=2$ kg m^{-3} and $\rho_0=1025$ kg m^{-3}, $c=0.85$ m s^{-1}.

2. Why internal waves exist in a stratified ocean

$$N^2 = -(g/\rho_0)\, d\rho/dz > 0$$

Here z is positive upward. In stable stratification, density increases downward, so $d\rho/dz<0$ and $N^2>0$. If a parcel is displaced upward without immediately changing its density, it is denser than its new surroundings and sinks. If displaced downward, it is lighter and rises. Inertia carries it past equilibrium, producing an oscillation at a frequency related to the buoyancy frequency N.

Stability interpretation. $N^2>0$ supports small-amplitude internal oscillations. $N^2\approx0$ means weak restoring force in a nearly homogeneous layer. $N^2<0$ means gravitational instability: overturning convection grows rather than a stable small oscillation. With rotation, freely propagating internal-wave frequencies usually lie between the inertial frequency f and N.

Interfacial model

The idealized ocean has two nearly homogeneous layers separated by a thin density jump. Displacement is concentrated at the interface and the layers commonly move in opposite directions. Reduced gravity summarizes the small internal density contrast. This model is useful for a sharp pycnocline and can support nonlinear solitary waves.

Continuous stratification

Density and N vary smoothly with depth, so motion occupies a depth range instead of a single interface. Boundary conditions select vertical modes with nodes and phase changes; each baroclinic mode has its own speed and vertical structure. Propagating beams can also be described by the internal-wave dispersion relation. The first baroclinic mode generally has the largest vertical scale and fastest mode speed.

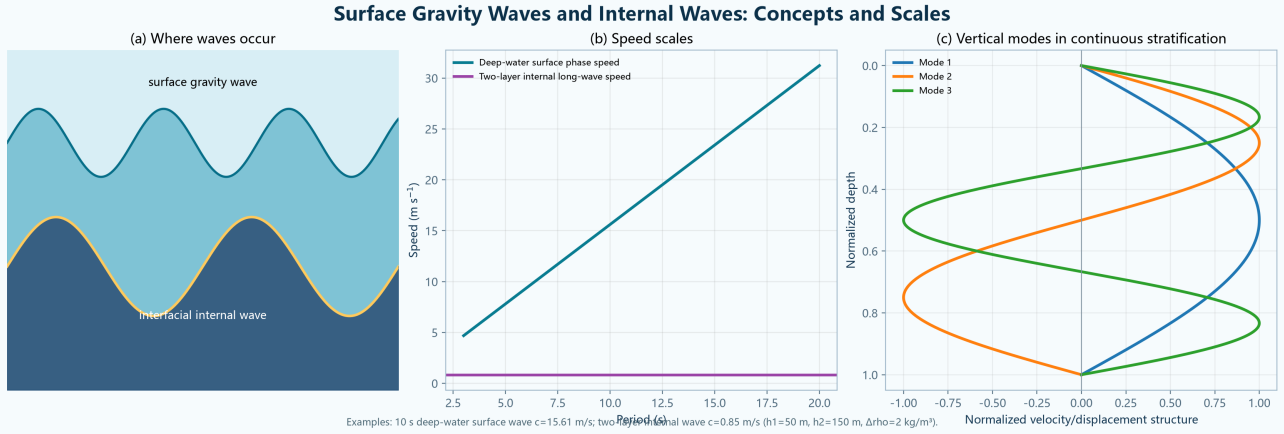


Figure 1. Locations, representative speed scales and idealized vertical modes. The comparison uses phase speed and is intended to show scale; real values depend on depth, wavelength and stratification.

3. Observational case study

The surface-wave record comes from NOAA NDBC station 44088 at the New York Harbor entrance. Of the 2024 observations, 17,197 contain valid significant wave height and dominant period. Median Hs is 1.31 m, the 95th percentile is 2.86 m and the maximum is 7.16 m. Hs is a statistical sea-state measure, approximately the mean height of the highest one third of waves, rather than the amplitude of one sinusoidal wave.

The internal-wave environment is diagnosed from an official Argo profile at 114.506 E, 15.542 N in the South China Sea on 16 March 2017. Adjusted pressure, temperature and salinity with quality flags 1 or 2 are preferred. TEOS-10 converts the observations to Absolute Salinity, Conservative Temperature, potential-density anomaly and N^2. Of the calculated intervals, 99.0% have $N^2>0$; the maximum N^2 corresponds to a minimum buoyancy period of 3.49 min.

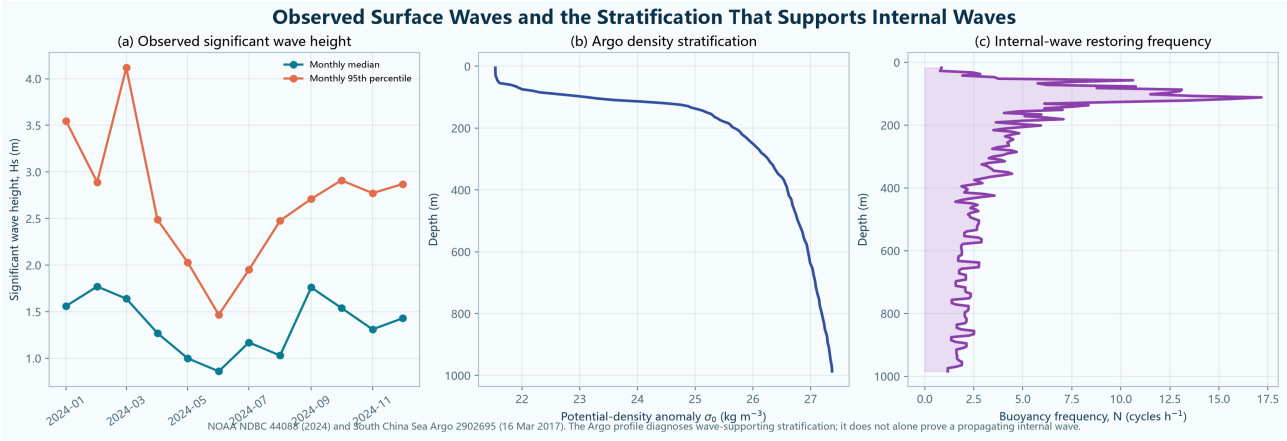


Figure 2. (a) Monthly surface-wave statistics from NDBC 44088. (b) South China Sea potential density. (c) Positive buoyancy frequency. The buoy directly observes surface-wave statistics. The Argo profile establishes wave-supporting stratification but does not alone prove propagation.

Observational limit. A propagating internal wave requires time-resolved evidence of frequency, wavelength and phase movement. Suitable observations include moored temperature chains and current meters, rapidly repeated CTD/glider sections, coordinated density and velocity measurements, or sequences of satellite images showing surface-roughness bands.

4. Submission-ready integrated answer

Ocean surface gravity waves are disturbances of the air-sea interface. Water parcels mainly move in orbital paths while the wave form and energy propagate. Gravity acting on the displaced free surface provides the restoring force. Their behavior depends on wavelength relative to water depth: deep-water waves are dispersive, with $\omega^2=gk$ and phase speed $c_p=\omega/k$, whereas shallow-water waves travel approximately at $c=(gH)^{1/2}$.

Internal waves are oscillations of density surfaces within a stably stratified ocean. They occur because a parcel displaced vertically from equilibrium has a different density from the surrounding water. Buoyancy accelerates it back; inertia causes overshoot and sustained oscillation. With z positive upward, stable stratification has $N^2=-(g/\rho_0)d\rho/dz>0$. In a rotating continuously stratified ocean, freely propagating internal waves usually satisfy $f<\omega<N$.

The two wave classes differ in at least four ways. First, surface waves occur at the free surface, while internal waves occur within the water column. Second, surface waves are restored by gravity acting across the large air-water density contrast, while internal waves are restored by buoyancy associated with much smaller seawater density differences. Third, surface gravity waves are generally faster; for example, a 10 s deep-water wave has $c_p\approx15.6$ m s^{-1}, whereas the example two-layer internal wave travels at about 0.85 m s^{-1}. Fourth, surface elevation is commonly centimetres to several metres, whereas internal isopycnal displacement can be tens of metres or more even when its surface expression is small.

In the interfacial model, two nearly homogeneous layers are separated by a sharp density jump. Displacement is concentrated at the interface, reduced gravity $g'=g\Delta\rho/\rho_0$ controls the restoring force, and the long-wave speed is $c=[g'h_1h_2/(h_1+h_2)]^{1/2}$. In a continuously stratified ocean, density and N vary smoothly with depth. Internal waves occupy a depth range and form vertical modes or beams; different baroclinic modes have different node patterns and propagation speeds. Thus the interfacial model is a useful simplified limit of a sharp pycnocline, while continuous stratification represents the broader vertical structure of the real ocean.

Concise conclusion

Both are gravity waves, but their relevant density boundary differs. The free surface produces fast, visible surface waves; weak internal density contrasts produce slower waves with much larger vertical displacement inside the ocean.

5. Quality control, data and references

Self-check

1. The response defines both wave classes and provides five explicit comparisons.
2. The stability sign convention is stated: z positive upward and $N^2>0$ for stable stratification.
3. Surface elevation is distinguished from isopycnal displacement.
4. The two-layer interface is distinguished from continuous vertical modes.
5. MATLAB independently reproduces $c_p=15.613100$ m s^{-1}, $\lambda=156.1310$ m and $c_i=0.847234$ m s^{-1}.
6. The Argo profile is presented as evidence of stable stratification, not proof of a propagating wave.

Data and methods

NOAA NDBC historical standard meteorological data: https://www.ndbc.noaa.gov/data/historical/stdmet/
Argo GDAC profile: https://data-argo.ifremer.fr/dac/csio/2902695/profiles/D2902695_039.nc
TEOS-10 gsw_Nsquared: https://www.teos-10.org/pubs/gsw/html/gsw_Nsquared.html

Selected references

Alford, M. H. et al. (2015). The formation and fate of internal waves in the South China Sea. *Nature*, 521, 65-69. https://doi.org/10.1038/nature14399

IOC, SCOR and IAPSO (2010). The International Thermodynamic Equation of Seawater - 2010: Calculation and Use of Thermodynamic Properties. UNESCO.

NOAA National Data Buoy Center. Wave observations and significant-wave-height documentation. https://www.ndbc.noaa.gov/faq/wavecalc.shtml

Reproducibility package

The downloadable ZIP contains both official raw files, processed CSV tables, a JSON summary with SHA-256 hashes, the Python analysis and figure generator, the MATLAB verification script, and a README describing the inference limit.